

## Ghosts and shadows- evidential contrafactuals in Georgian.

Factive predicates such as *know*, *remember*, *realize*, *etc* that presuppose the truth of their complements have been extensively discussed in the literature. There are also predicates such as *hallucinate*, *be wrong* that entail the falsity of their complements. However, it has been argued that natural language lacks so called *contrafactuals*, i.e belief predicates that presuppose the falsity of their complements rather than their truth (Holton, 2017), which are essentially the opposite of *know*. This, however, has been called into question by a few cross-linguistic counterexamples like the Spanish predicate *se creerse* (Anvari et al., 2019), and the Mandarin predicate *yǐwéi* (Glass, 2023). In this paper, I present novel data on the Georgian predicate *mo-e-čven-a* as another apparent counterexample to this claim.<sup>1</sup> I argue that *mo-e-čven-a* has a semantics similar to that of *se creerse*, with an added requirement of direct evidentiality.

A true contrafactive must apparently satisfy the following criteria: 1) Stative. 2) Monomorphemic 3) Doxastic (subject believes the complement) 4) contrafactive (presupposes the negation of the complement) (Holton, 2017; Roberts and Özyıldız, 2025). Of these, *mo-e-čven-a* satisfies three of the diagnostics. It is stative, doxastic, and contrafactive. While it is not monomorphemic, verbal predicates in Georgian typically combine with different preverbs and pre-radical vowels, giving rise of a wide range of meanings and do not occur as just roots. The root in this case *čven* only has the semantics of false belief when combined with the preverb *mo-* and the pre-radical vowel *-e-*.

*mo-e-čven-a* gives rise to false belief inferences in a simple declarative (1a). Unlike the predicate *imagine*, (1a) requires that the attitude holder believe that it was raining, when it was not actually raining. As in the case of the predicate *se creerse*, the false belief inference seems to project from under questions as in (1b) but not from negation. We see that with negation in (1c) the inference is that the proposition is true (*p*) rather than false ( $\neg p$ ), while under questions as in (1b)  $\neg p$  seems to project, which suggests that it may be a presupposition. Furthermore, unlike *se creerse*, it can also be used in the first person in the positive form, and still generate a false belief inference (1d).

These characteristics of *mo-e-čven-a* can be contrasted with the Georgian predicate *e-gon-a* in (2) which can also be used to ascribe false belief but gives rise to weaker falsity inferences, but is doxastically neutral much like the English predicate *believe*. One interesting characteristic that sets *mo-e-čven-a* apart from other contrafactuals of this sort, is that it can only be used to report events that the attitude holder has directly perceived, or in this case, incorrectly perceived an event. But it cannot be used to report beliefs that are not based on direct perception of events as in (1e). I propose a decomposition of *mo-e-čven-a* where *mo-* encodes the false belief inference as a presupposition (3a), while *čven* encodes direct evidence (3b).

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<sup>1</sup>Unless otherwise specified, the data presented here was collected and independently verified from 5 Georgian native speakers in Tbilisi, Georgia.

- (1) a. nino-s mo-e-čven-a rom c'vim-d-a  
 nino-DAT PVB-PV-evidence-AOR.3.SG COMP rain-IMPf-AOR.3.SG  
 Nino [falsely] believed that it was raining. ( ~ *it was not raining.* )
- b. nino-s mo-e-čven-a rom c'vim-d-a [tu ara]?  
 nino-DAT PVB-PV-evidence-AOR.3.SG COMP rain-IMPf-AOR.3.SG [POL]?  
 Did Nino [falsely] believe that it was raining? ( ~ *it was not raining.* )
- c. nino-s ar mo-e-čven-a rom c'vim-d-a. martla  
 nino-DAT NEG PVB-PV-evidence-AOR.3.SG COMP rain-IMPf-AOR.3.SG. really  
 c'vim-d-a.  
 rain-IMPf-AOR.3.SG  
 Nino did not [falsely] believe that it was raining. It was really raining.
- d. mo-m-e-čven-a rom c'q'ali  
 PVB-PART.1.SGPV-evidence-AOR.3.SG COMP water.NOM  
 dal-i-e  
 drink-THEM.SUFF-PART.2.SG.  
 I [falsely] believed that you drank water. ( ~ *You did not drink water.* )
- e. q'vela st'udent'-s mo-e-čven-a rom ✓c'vim-d-a/  
 Every student-DAT PVB-PV-evidence-AOR.3.SG COMP rain-IMPf-AOR.3.SG/  
 ✗gamocd-eb-i ča-a-bar-es.  
 exam-PL-NOM PVB-PV-pass-AOR.3.PL.  
 Every student [falsely] believed that it was raining/they passed the exam.
- (2) nino-s e-gon-a rom c'vim-d-a  
 nino-DAT PV-believe-AOR.3.SG COMP rain-IMPf-AOR.3.SG  
 Nino believed that was raining.
- (3) a.  $\llbracket mo- \rrbracket(P) = \lambda x. \lambda w : P(x)(w)$  is defined and  $B_p(w) = 0$  (Adopting the denotation for WRONGLY in Maldonado and Percus (2024))
- b.  $\llbracket cven(f)(B)(w)(\phi) \rrbracket = 1$  is only defined if for all worlds  $w', w' \in B(w)$   
*iff* the perceived evidence in  $w$  holds in  $w'$  and  $f$  is a choice function of type  $\langle st, st \rangle$   
 such that  $f(B(w)) \subseteq B(w)$ .  
 If defined,  $\llbracket cven(f)(B)(w)(\phi) \rrbracket = 1$  *iff* for  $\forall w' \in f(B(w)) : \llbracket \phi(w') \rrbracket = 1$   
 (Adopting the denotation for a direct evidential from Matthewson et al. (2006)  
 for St'at'imcets)

## References.

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